

## NEUBURG, Germany

WMO: 108530

Lat: 48.7086N

Lon: 11.2147E

Elev: 1300

StdP: 14.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.9        | 15.5 | 6.7                           | 8.2 | 11.3 | 12.3 | 10.8 | 17.6 | 26.5                  | 43.9 | 23.0 | 41.9 | 3.3                   | 40   | 0.515 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.6                   | 87.5            | 67.4 | 83.9 | 66.3 | 80.7 | 65.0 | 69.0                | 82.9 | 67.5 | 80.4 | 66.1 | 78.0 | 5.4                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.5                            | 95.3 | 73.6 | 62.9 | 90.1 | 72.0 | 61.6 | 86.0 | 71.0 | 33.9          | 83.2 | 32.7 | 80.4 | 31.5 | 78.0 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |       |     |
| (4) 19.6          | 16.4 | 13.8 | DB                         | 3.3 | 92.6               | 6.7 | 3.0                                                | -1.5 | 94.7       | -5.5 | 96.4       | -9.2 | 98.1       | -14.1 | 100.3 | (4) |
| (5)               |      |      | WB                         | 2.7 | 71.8               | 6.7 | 1.6                                                | -2.1 | 73.0       | -6.0 | 73.9       | -9.7 | 74.8       | -14.5 | 75.9  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 48.8   | 31.1  | 33.7  | 40.9  | 48.8  | 56.9  | 63.3  | 66.1  | 65.6  | 57.3  | 48.4  | 39.6  | 33.2  | (6)  |  |
|                                                                        | DBStd       | 14.19  | 8.32  | 8.32  | 7.09  | 6.92  | 6.37  | 6.41  | 5.68  | 5.70  | 5.93  | 6.47  | 6.41  | 7.69  | (7)  |  |
|                                                                        | HDD50       | 2400   | 585   | 456   | 288   | 103   | 14    | 1     | 0     | 0     | 8     | 109   | 316   | 520   | (8)  |  |
|                                                                        | HDD65       | 6145   | 1049  | 875   | 746   | 486   | 260   | 108   | 57    | 64    | 239   | 514   | 762   | 985   | (9)  |  |
|                                                                        | CDD50       | 1975   | 1     | 0     | 6     | 67    | 229   | 399   | 498   | 484   | 227   | 60    | 4     | 0     | (10) |  |
|                                                                        | CDD65       | 248    | 0     | 0     | 0     | 0     | 10    | 57    | 90    | 83    | 8     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 2845   | 0     | 0     | 0     | 18    | 179   | 656   | 975   | 885   | 129   | 3     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 829    | 0     | 0     | 0     | 1     | 28    | 180   | 316   | 288   | 16    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.4    | 6.0   | 6.0   | 6.7   | 5.8   | 5.5   | 5.0   | 5.2   | 4.7   | 4.6   | 4.7   | 5.1   | 5.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 30.6   | 2.0   | 1.6   | 2.0   | 1.9   | 3.1   | 3.5   | 3.7   | 3.2   | 2.5   | 2.3   | 2.4   | 2.3   | (15) |  |
|                                                                        | PrecMax     | 40.8   | 5.2   | 4.0   | 6.0   | 5.6   | 5.7   | 5.9   | 6.7   | 5.8   | 5.3   | 7.7   | 5.1   | 5.1   | (16) |  |
|                                                                        | PrecMin     | 20.5   | 0.1   | 0.6   | 0.6   | 0.4   | 0.4   | 1.3   | 1.0   | 1.0   | 0.6   | 0.5   | 0.0   | 0.2   | (17) |  |
|                                                                        | PrecStd     | 4.3    | 1.1   | 0.8   | 1.3   | 1.1   | 1.4   | 1.1   | 1.4   | 1.4   | 1.3   | 1.5   | 1.1   | 1.1   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 53.7  | 58.0  | 66.1  | 76.6  | 83.5  | 89.5  | 92.0  | 92.6  | 82.4  | 72.1  | 60.6  | 53.4  | (19) |  |
|                                                                        |             | MCWB   | 47.9  | 48.3  | 51.7  | 57.4  | 64.1  | 69.4  | 68.4  | 68.1  | 64.5  | 59.4  | 54.1  | 48.5  | (20) |  |
|                                                                        | 2%          | DB     | 49.9  | 51.9  | 61.0  | 71.7  | 79.1  | 85.0  | 87.5  | 86.8  | 78.0  | 66.5  | 55.5  | 49.7  | (21) |  |
|                                                                        |             | MCWB   | 45.4  | 45.5  | 49.6  | 55.5  | 62.5  | 67.1  | 67.2  | 67.5  | 63.3  | 58.1  | 50.6  | 45.8  | (22) |  |
|                                                                        | 5%          | DB     | 45.9  | 48.7  | 56.7  | 67.5  | 74.9  | 81.3  | 83.3  | 82.7  | 73.7  | 62.7  | 51.9  | 46.6  | (23) |  |
|                                                                        |             | MCWB   | 42.4  | 43.8  | 47.7  | 54.1  | 60.1  | 65.3  | 66.0  | 66.0  | 61.9  | 56.1  | 48.4  | 43.6  | (24) |  |
|                                                                        | 10%         | DB     | 42.4  | 45.0  | 52.4  | 62.9  | 70.6  | 77.4  | 79.4  | 79.0  | 69.7  | 58.9  | 48.9  | 43.2  | (25) |  |
|                                                                        |             | MCWB   | 39.9  | 41.1  | 46.0  | 51.7  | 57.8  | 63.6  | 64.8  | 64.6  | 59.7  | 54.1  | 46.1  | 41.1  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 49.0  | 49.1  | 53.3  | 59.7  | 66.1  | 71.2  | 70.6  | 70.9  | 66.9  | 61.5  | 54.0  | 49.3  | (27) |  |
|                                                                        |             | MCDB   | 52.6  | 55.8  | 62.3  | 73.2  | 79.6  | 85.3  | 86.1  | 84.8  | 77.7  | 68.8  | 59.0  | 52.4  | (28) |  |
|                                                                        | 2%          | WB     | 45.2  | 46.1  | 50.7  | 56.7  | 63.4  | 68.5  | 68.6  | 68.8  | 64.6  | 58.9  | 51.1  | 46.4  | (29) |  |
|                                                                        |             | MCDB   | 49.1  | 51.8  | 58.9  | 68.9  | 76.4  | 81.3  | 82.7  | 82.6  | 75.5  | 65.1  | 54.6  | 49.7  | (30) |  |
|                                                                        | 5%          | WB     | 42.3  | 43.8  | 48.6  | 54.6  | 61.2  | 66.6  | 67.3  | 67.2  | 62.8  | 56.9  | 48.8  | 43.5  | (31) |  |
|                                                                        |             | MCDB   | 45.3  | 48.2  | 55.3  | 65.3  | 72.8  | 78.7  | 80.2  | 80.0  | 71.7  | 62.0  | 51.6  | 46.1  | (32) |  |
|                                                                        | 10%         | WB     | 39.7  | 41.3  | 46.4  | 52.5  | 59.1  | 64.7  | 65.7  | 65.5  | 60.9  | 54.5  | 46.3  | 41.1  | (33) |  |
|                                                                        |             | MCDB   | 42.0  | 44.9  | 52.0  | 62.0  | 68.8  | 75.7  | 76.9  | 76.9  | 68.3  | 58.4  | 48.6  | 43.1  | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.7   | 13.4  | 16.5  | 20.0  | 19.7  | 20.4  | 20.6  | 20.9  | 19.6  | 16.0  | 10.5  | 9.2   | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 14.2  | 19.4  | 26.0  | 29.2  | 28.1  | 27.8  | 28.4  | 28.6  | 27.3  | 22.5  | 16.9  | 13.2  | (36) |  |
|                                                                        |             | MCWBR  | 10.7  | 13.3  | 15.8  | 15.2  | 13.3  | 12.5  | 11.4  | 11.8  | 14.2  | 14.2  | 12.4  | 10.2  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 13.6  | 18.6  | 23.5  | 26.6  | 25.3  | 25.3  | 24.8  | 25.8  | 23.9  | 20.9  | 15.7  | 12.8  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.303 | 0.326 | 0.360 | 0.392 | 0.401 | 0.405 | 0.410 | 0.404 | 0.379 | 0.353 | 0.325 | 0.299 | (40) |  |
|                                                                        | taud        |        | 2.439 | 2.381 | 2.328 | 2.268 | 2.274 | 2.294 | 2.273 | 2.311 | 2.358 | 2.423 | 2.461 | 2.451 | (41) |  |
|                                                                        | Ebn at Noon |        | 243   | 261   | 269   | 271   | 273   | 272   | 268   | 265   | 260   | 248   | 231   | 228   | (42) |  |
|                                                                        | Edh at Noon |        | 20    | 27    | 33    | 39    | 41    | 40    | 40    | 37    | 32    | 26    | 20    | 18    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 301   | 570   | 932   | 1365  | 1598  | 1787  | 1700  | 1483  | 1066  | 648   | 333   | 240   | (44) |  |
|                                                                        | RadStd      |        | 43    | 82    | 119   | 181   | 182   | 147   | 152   | 142   | 131   | 78    | 41    | 29    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | +0.68 | +1.46 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (62 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | +0.63 | N/A         | N/A   | +112  | +41   |

Nomenclature: See separate page